Real-Time Crowd Classification Using Deep Convolutional Neural Networks with Data Augmentation for Enhanced Public Safety

Saksham Sharma[1, a)], Disha Saini[1, b)], Aman Singh[1, c)] and Kartik Vats[1, d)]

[1] National Institute of Electronics & Information Technology, Ropar, Punjab, India
[a)] saksham.sharma01123@gmail.com
[b)] d1shaa0311@gmail.com
[c)] amansingh9327@gmail.com
[d)] kartikluckyvats@gmail.com

Abstract. Crowd management is one of the most important aspects of human society, aimed at protecting the lives of citizens. This study presents a comprehensive evaluation of Convolutional Neural Network (CNN) architectures for classifying images into Crowded and Uncrowded categories. Among the non-augmented models, the three-layer CNN (CNN3) achieves the highest accuracy (0.82), macro F1-score (0.82), and AUC (0.88), demonstrating superior discriminatory capability. Data augmentation improves robustness, particularly for the Crowded class, with CNN2 + Augmentation achieving a recall of 0.87 and CNN3 + Augmentation maintaining a strong AUC of 0.82. While augmentation slightly reduces peak accuracy, it enhances class balance and generalisation across the datasets. ROC curve analysis further confirms the superior performance of deeper architectures, with three-layer Sigmoid-based models outperforming two-layer counterparts. Overall, the results highlight that the three-layer CNN with Sigmoid activation provides a robust baseline, and augmented versions deliver enhanced generalisation and class-balanced performance, making them well-suited for practical real-time crowd-classification applications.

Keywords: Crowd Management, CNN, Image Processing, Data Augmentation, Public Safety, IoT.

1 Introduction

Crowds have been an integral part of human civilisation since the earliest gatherings for trade, worship, celebrations, and political discourse. The earliest recorded instances of large crowds date back to ancient civilisations such as Mesopotamia, Egypt, and Rome, where public festivals, markets, and coliseum events attracted thousands of people at a time. Although crowds were historically viewed as natural outcomes of social and economic activities, their behaviour patterns were not systematically studied until the late nineteenth century.

The concept of crowd management first emerged in the late 1800s and early 1900s, when rapid urbanisation and industrial growth led to densely populated cities and

frequent mass gatherings. The tragic incidents during early sporting events, religious gatherings, and political movements highlighted the need for structured approaches to manage human flow, prevent panic, and ensure public safety. In the modern era, crowd management has evolved into a multidisciplinary field involving urban planning, behavioural science, safety engineering, and, more recently, artificial intelligence.

Today, population concentration in urban centres has intensified crowd-related challenges. Several megacities around the world face issues such as pedestrian congestion, transportation overload, and crowd surges during events. The table below presents five of the most crowded cities globally, considering population density and urban area.

Table 1. World's Top ten Most Crowded Cities.

Rank	City	Country	Urban Area (sq. km)	Population Density (people/sq. km)
1	Manila	Philippines	~150	~41,515
2	Dhaka	Bangladesh	~306	~30,093
3	Karachi	Pakistan	~1,100	~24,000
4	Mumbai	India	~603	~20,634
5	Kolkata	India	~205	~24,252
6	Lagos	Nigeria	~1,171	~7,941
7	Shenzhen	China	~1,997	~6,600
8	Seoul	South Korea	~605	~16,028
9	Jakarta	Indonesia	~664	~14,464
10	Tokyo (central wards)	Japan	~622	~14,650

With the growing scale of urban crowds, traditional manual monitoring methods are no longer sufficient. This has paved the way for AI-based crowd detection and management systems using machine learning and computer vision. Modern systems analyse CCTV feeds, drone footage, and sensor data to estimate crowd density, identify abnormal behaviour, detect bottlenecks, and predict potential stampede risks in real time. Techniques such as Convolutional Neural Networks (CNNs), YOLO-based object detection, density-map estimation, and human-flow analysis are increasingly being deployed in smart cities, transport hubs, sports arenas, and religious gatherings. Some commercial solutions—such as CrowdVision, VaakAI, Numina, and iOmniscient's iQ-Crowd—already provide advanced automated crowd analytics for public safety and urban mobility planning. In this context, developing reliable machine learning-based

crowd detection models is crucial for enhancing safety, preventing disasters, and supporting the future of intelligent urban management.

2 Literature Review

The study [16] presents real-time methods for addressing challenges associated with dense crowds. It utilises static images captured from crowded areas to estimate crowd density using two distinct approaches: traditional image processing and Convolutional Neural Networks (CNNs). In the image processing method, erosion is applied to identify and count head regions within grayscale images. The CNN-based approach leverages deep learning to predict crowd density more accurately. Additionally, a network of Raspberry Pi devices is employed to carry out crowd detection and relay real-time crowd information to the relevant authorities. This enables officials to be promptly alerted to potential stampede risks and take preventive measures to ensure public safety.
The study [15] focuses on managing crowd congestion to enhance safety and prevent stampedes. Hajj has been selected as the case study, and an E-wrist belt integrating multiple sensors with a Wemos D1 module via the Internet of Things (IoT) is proposed. Stampedes in large gatherings often occur due to fluctuations in attendance, temperature, and humidity. A machine learning approach is employed to estimate the likelihood of stampedes by analyzing these factors. The primary aim of this research is to enhance the existing crowd management system during Hajj through the implementation of the E-wrist belt.
The study [7] proposes an IoT-based system in which vibration sensors are installed beneath a platform to measure vibrations generated by people walking on it. When these vibrations exceed a predefined threshold, indicating a higher risk of crowding, an alert is sent to the authorities. The data collected from the sensors can be analyzed to identify patterns and predict potential crowding or stampede events. To validate the system, a prototype will be constructed using an aluminium or wooden plank, with vibrations manually induced using motors to test the IoT circuit. The ultimate goal is to deploy this solution in high-risk areas to prevent injuries and fatalities caused by stampedes.
The systematic literature review (SLR) [3] examines crowd management in modern societies, focusing on both global and local contexts, including Hajj events in Saudi Arabia, through the lens of deep learning (DL) techniques. This survey covers research published between 2010 and 2023 and identifies 45 primary studies that address the research questions, exploring taxonomies, methodologies, and comprehensive approaches in crowd management. Both supervised and unsupervised DL techniques were found to achieve high accuracy, each with distinct strengths and limitations. Many studies emphasize scene analysis of crowds captured via installed cameras, but the potential use of social media data in crowd analysis remains a challenge. Leveraging big data could significantly advance crowd management strategies. Motivated by these findings, the review aims to highlight the key obstacles and dilemmas in the field, provide a roadmap for future research, and identify existing gaps, particularly the limited studies on Hajj, sentiment analysis, and the behavior of pilgrims.

The study [1] proposes a conceptual framework aimed at optimizing risk management in the events sector through advanced technologies, including artificial intelligence (AI), the Internet of Things (IoT), and real-time analytics. Using a User-Centred Design (UCD) approach and Action Research Methodology, the study gathers insights from key stakeholders such as risk managers, event organizers, and technology experts to identify user requirements and preferences. The findings highlight the importance of collaborative tools, AI-driven decision-making, customizable features, and dual-format accessibility in fostering proactive risk management. Although the framework is not fully operational, it provides practical recommendations for developing innovative solutions that enhance safety, efficiency, and inclusivity in event planning and execution. This framework lays the groundwork for future advancements, addressing real-world challenges while supporting Saudi Arabia's vision of becoming a global leader in the events industry.

Crowd management [12] plays a vital role in ensuring public safety and optimizing space utilization in areas with high pedestrian traffic, such as urban centers, sports stadiums, transportation hubs, and large-scale events. Traditional methods often rely on manual surveillance and estimations, which pose challenges for real-time decision-making and risk mitigation. The emergence of machine learning (ML) offers new opportunities to enhance crowd management strategies. This study examines the application of ML techniques for tasks such as crowd density estimation, flow prediction, behavior analysis, and risk assessment. The findings indicate that ML-based systems can substantially improve the accuracy and efficiency of crowd monitoring and control by providing real-time analytics, enhanced situational awareness, and predictive capabilities, enabling more informed decisions by authorities. The paper also highlights implementation challenges, including data privacy, the need for extensive and diverse datasets, and ensuring system robustness. Future directions include integrating advanced ML methods like deep learning, developing real-time adaptive learning systems for dynamic crowd management, and addressing ethical and regulatory considerations. Leveraging ML can transform crowd management into a proactive, data-driven, and responsive practice, significantly enhancing public safety and space management.

The Integrated Intelligent Crowd Control and Management (IICCM) [2] framework leverages advanced technologies, including Computer Vision (CV), Artificial Intelligence (AI), and the Internet of Things (IoT), to enhance participant safety and optimize crowd management. CV provides accurate real-time detection and tracking, AI analyzes crowd behavior to anticipate potential risks, and IoT collects environmental data to improve crowd flow, reduce congestion, and enable timely interventions. The framework also supports emergency evacuation planning by simulating crowd dynamics and identifying safe and efficient escape routes. While applicable to a variety of events, the Hajj pilgrimage—a large and dynamic annual gathering—serves as a rigorous test case, demonstrating the system's adaptability and robustness in managing millions of participants from diverse cultural and linguistic backgrounds. By successfully addressing the challenges specific to Hajj, the IICCM framework proves its scalability and potential for deployment in other large-scale events, providing valuable guidance for decision-makers aiming to implement advanced crowd management solutions.

The COVID-19 pandemic necessitated strict restrictions at crowded events [14] to limit the spread of the virus. Despite these measures, many events failed to comply, becoming super-spreader occurrences. To control the transmission of human-to-human communicable diseases, measures such as wearing face masks, maintaining social distancing, and following regular cleaning and sanitization protocols are essential, particularly at crowded gatherings. Some events occur spontaneously, including political rallies, protest marches, or funeral processions, making them especially challenging to manage during a crisis like COVID-19. This study reviews several notable crowded events during the pandemic and proposes a machine learning-based framework to effectively organize such events during ongoing and future crises. The framework includes validation metrics for its components and an extensive algorithm, with explanations of its functionalities. Moreover, the proposed system can be adapted for managing events in other crisis scenarios.

Urban environments in India are characterized by dense crowds, cultural diversity, and complex work settings, which pose unique challenges for crowd management, crime prevention, and workplace supervision [6]. This study proposes a customized framework that leverages Artificial Intelligence (AI) and Machine Learning (ML) to address these challenges in real time. In crowded public spaces, where chaotic dynamics and culturally specific activities can create risks, the framework employs advanced computer vision to monitor and analyze crowd behavior, enabling authorities to detect anomalies quickly and ensure public safety. Machine learning-driven predictive analytics anticipate potential issues by examining historical data and real-time trends, facilitating culturally sensitive preventive measures. For crime prevention, AI algorithms analyze crowd patterns to identify individuals with known criminal histories, allowing law enforcement to take proactive measures and enhance security. In industrial environments, the framework uses AI-powered surveillance and data analytics to monitor employee activities, enforce safety protocols, and optimize workflow efficiency. By integrating these technologies, the framework offers a comprehensive solution to improve safety and operational efficiency across India's diverse urban settings, promoting a secure and culturally aware environment.

The concept of smart cities has advanced significantly due to the development of technologies such as mobile edge computing (MEC), ultra-dense networking, and software-defined networking [17]. However, designing efficient routing strategies becomes increasingly complex due to the dynamic distribution of crowds across various sectors of a smart city, which imposes stringent and constantly changing network requirements. To address network congestion and balance the load for smart city services, we propose a deep reinforcement learning-based smart routing algorithm. This approach ensures the distributed computing and communication infrastructure operates efficiently while meeting the latency requirements of service requests from the crowd. The study also presents extensive numerical results to demonstrate and validate the effectiveness of the proposed algorithm.

In the current scenario, numerous events such as political rallies, live concerts, strikes, and sports gatherings attract large crowds [13]. In such crowded settings, the likelihood of suspicious activities and violent incidents increases. To mitigate these risks, it is crucial to develop automated systems capable of detecting anomalies in complex

outdoor environments, particularly in public spaces, to ensure safety and prevent crowd-related disasters such as stampedes, mob violence, and riots. Effective crowd management can help avert such incidents and maintain public safety in locations including temples, railway stations, airports, bus terminals, religious gatherings, and political events. A visual surveillance system that can automatically detect abnormal crowd behavior is essential for enabling timely interventions. Core steps in crowd analysis include density estimation, crowd counting, object recognition, tracking, and anomaly detection. This study systematically reviews and compares various methods used in crowd analysis, presenting a comparative evaluation of existing approaches across multiple datasets and analyzing different taxonomies.

The reviewed literature demonstrates a range of approaches for crowd monitoring and management, from sensor-based IoT solutions and wearable devices to machine learning-driven predictive frameworks. While traditional methods, including manual surveillance and basic image processing, provide foundational insights, they often lack the scalability, real-time responsiveness, and robustness required for dynamic crowd scenarios. Deep learning, particularly Convolutional Neural Networks (CNNs), has shown significant potential in addressing these limitations by automatically learning complex spatial features from images, enabling accurate crowd density estimation, classification, and anomaly detection. Given the growing demand for real-time monitoring in large-scale events, urban public spaces, and emergency scenarios, there is a clear need to develop CNN-based models capable of classifying images into crowd and uncrowd categories. Such models not only enhance situational awareness but also empower authorities to implement timely interventions, prevent overcrowding, and mitigate risks of stampedes or other crowd-related hazards. Therefore, integrating CNN techniques into crowd management systems represents a critical step toward more proactive, intelligent, and reliable public safety solutions.

3 Proposed Methodology for Crowd Level Detection Using CNN

Fig. 1 shows the complete workflow for classifying crowd levels using a Convolutional Neural Network (CNN) framework. The process begins with dataset creation using Kumbh 2025 images, labelled into two categories: Overcrowded and Normal Crowd. The images are normalised and enhanced through several augmentation techniques, including rotation, contrast, hue, saturation adjustments, and noise addition, followed by splitting the dataset into an 80:20 ratio. Two CNN architectures are designed: a baseline model and a modified model that incorporates Batch Normalisation and higher dropout rates for better regularisation. Both models are evaluated using training accuracy and loss curves, along with confusion-matrix-based performance metrics such as precision, recall, and F1-score. A final evaluation step determines whether the model's performance meets the required criteria; if successful, the workflow concludes with the validated model.

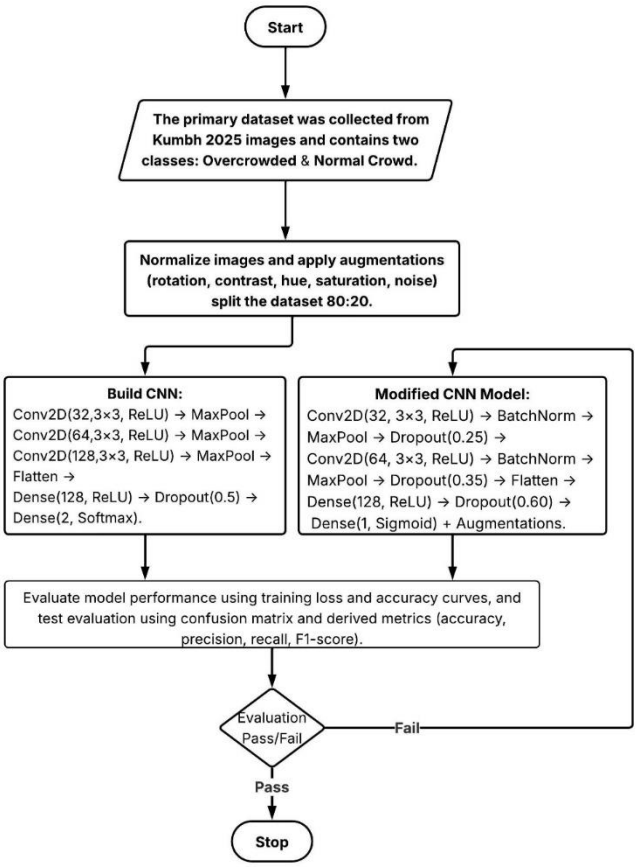

Fig. 1. Proposed Methodology of the CNN-Based Crowd Classification.

3.1 Data Preprocessing

The dataset images were collected from multiple sources, including newspapers, magazines, Google Images, blogs, and YouTube channels. The images were organised into two separate folders: one containing crowded images and the other containing uncrowded images of Kumbh Mela. Large-sized images were resized to a uniform dimension of 150×150×3 pixels to make them suitable for computational processing. To facilitate efficient training, all image pixel values were normalised to a range between 0 and 1 (see Eq. 1). Basic data augmentation techniques were applied to enhance model generalisation, including rotation (see Eq. 2-3), hue adjustment, contrast adjustment, saturation reduction (see Eq. 4-6), and the addition of Gaussian noise. Finally, the dataset was split into training and testing sets in an 80:20 ratio.

$$X_{\text{norm}} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

Where: X is the original pixel value, $X_{\min}$ is the minimum pixel value in the dataset (usually 0), $X_{\max}$ is the maximum pixel value in the dataset (usually 255), and $X_{\text{norm}}$is the normalised pixel value.

$$x' = xCos\theta - ySin\theta \quad (2)$$

$$y' = xSin\theta + yCos\theta \quad (3)$$

Where the original coordinates of the points are (x, y) and (x', y') are after rotation. θ is the angle of rotation (counterclockwise if $\theta > 0$).

$$(R', G', B') = f^{-1}\big((f(R,G,B)_H + \Delta H)\ mod\ 1,\ f(R,G,B)_S,\ f(R,G,B)_V\big) \quad (5)$$

$$I'(x,y) = \alpha \cdot (I(x,y) - m) + m \quad (6)$$

Where $I(x, y)$ is the original pixel intensity (0–255 for 8-bit images), $I'(x, y)$is the contrast-adjusted pixel intensity, $\alpha > 1$increases contrast, $0 < \alpha < 1$ decreases contrast, 128 is the midpoint of the intensity range (for 8-bit images).

$$I'(x,y) = I(x,y) + N(x,y), \quad N(x,y) \sim \mathcal{N}(\mu, \sigma^2) \quad (7)$$

Where: $I(x, y)$is the original pixel intensity, $I'(x, y)$ is the noisy pixel intensity, $N(x, y) \sim \mathcal{N}(\mu, \sigma^2)$is a random value drawn from a Gaussian distribution with mean μ (usually 0) and variance σ^2.

3.2 Proposed CNN Model Architecture

Table 2 shows the proposed CNN model [4], which consists of a series of convolutional, normalisation, pooling, dropout, flattening, and dense layers designed for feature extraction and classification. It begins with a Conv2D layer with 32 filters and a kernel size of 3×3, producing an output of shape (150, 150, 32) and 896 parameters. This is followed by Batch Normalisation (128 parameters) to stabilise and accelerate training. A MaxPooling2D layer reduces the spatial dimensions to (75, 75, 32), followed by a Dropout layer to prevent overfitting. The second convolutional block has 64 filters, producing an output of shape (75, 75, 64) with 18,496 parameters, followed by batch normalisation (256 parameters), max pooling to (37, 37, 64), and dropout. The Flatten layer converts the 3D feature maps into a 1D vector of length 87,616, which feeds into a Dense layer with 128 units and 11,214,976 parameters. A dropout layer is applied again before the final Dense output layer with 1 neuron for classification, having 129 parameters. In total, the model has 11,234,881 parameters, of which 11,234,689 are trainable, and 192 are non-trainable, reflecting a moderately complex network suitable for image classification tasks.

Table 2. Architecture of proposed CNN model.

Layer (type)	Output Shape	Param #
conv2d_6 (Conv2D)	(None, 150, 150, 32)	896
batch_normalization_8 (BatchNormalization)	(None, 150, 150, 32)	128
max_pooling2d_6 (MaxPooling2D)	(None, 75, 75, 32)	0
dropout_8 (Dropout)	(None, 75, 75, 32)	0
conv2d_7 (Conv2D)	(None, 75, 75, 64)	18,496
batch_normalization_9 (BatchNormalization)	(None, 75, 75, 64)	256
max_pooling2d_7 (MaxPooling2D)	(None, 37, 37, 64)	0
dropout_9 (Dropout)	(None, 37, 37, 64)	0
flatten_2 (Flatten)	(None, 87616)	0
dense_4 (Dense)	(None, 128)	11,214,976
dropout_10 (Dropout)	(None, 128)	0
dense_5 (Dense)	(None, 1)	129
Total params		11,234,881
Trainable params		11,234,689
Non-trainable params		192

In this paper, we begin with a simple three-layer convolutional neural network model. The architecture starts with a convolutional layer (see Eq. 8) containing 32 filters of size 3×3 with ReLU activation function (see Eq. 9), followed by a 2×2 max-pooling layer to reduce spatial dimensions (see Eq. 10). This is followed by a second convolutional layer with 64 filters and another max-pooling layer, enabling the model to extract higher-level image features. The feature maps are then flattened and passed to a dense layer with 128 neurons activated by ReLU, after which a dropout rate of 0.5 is applied to reduce overfitting. The final classification is performed by an output layer with two neurons and a Softmax activation function (see Eq. 11), making the model suitable for binary classification tasks. After training and evaluating the model, if signs of overfitting are observed, we incorporate necessary modifications, including data augmentation techniques such as rotation (30°), width and height shifts by 20% (see Eq. 12) [20], zoom augmentation by 30% (see Eq. 13), shear transformation by 15% (see Eq. 14), horizontal flipping (see Eq 15), brightness variation from 0.6 to 1.4 (see Eq. 16), and nearest-fill interpolation (see Eq, 17). Additionally, the output activation is changed from Softmax to the Sigmoid function (see Eq. 18) to better align with binary classification and improve model stability. These enhancements help improve generalisation and reduce overfitting, resulting in a more robust model[10].

$$(I * K)(x, y) = \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} I(x + i,\ y + j) \cdot K(i, j) \tag{8}$$

Where $I(x, y) \in R^{H \times W}$ is an input matrix, $K(i, j) \in R^{m \times n}$ is a filter matrix.

$$\mathrm{ReLU}(x) = \max(0, x) \tag{9}$$

$$Y(x, y) = \max_{0 \le i < m \ , \ 0 \le j < n} \quad I(x + i, \ y + j) \tag{10}$$

$$\mathrm{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{k=1}^{K} e^{z_k}} \tag{11}$$

$$(x'y') = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0.2H \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \tag{12}$$

Where H is the image height.

$$(x'y') = \begin{pmatrix} 1.3 & 0 & 0 \\ 0 & 1.3 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \tag{13}$$

$$(x'y') = \begin{pmatrix} 1 & 0.15 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \tag{14}$$

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & W \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \tag{15}$$

$$I'(x, y) = \beta \cdot I(x, y) \ where \ \beta \text{ is in range of } \ 0.6 \le \beta \le 1.4 \tag{16}$$

$$I'(x', y') = \mathrm{I}\big(round(x), \ round(y)\big) \text{ where } x = \frac{x'}{s_x}, y = \frac{y'}{s_y} \tag{17}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{18}$$

3.3 Optimisation and Gradient-Based Learning

The proposed model relies heavily on mathematical optimisation techniques to adjust their parameters so that predictions become increasingly accurate over time. Central to this process is the computation of loss, which quantifies the error between the predicted output and the true class label. Gradients of this loss function guide the model during training by indicating how the weights should change to minimise error. Algorithms such as Gradient Descent and its advanced variants, including the Adam optimiser, ensure stable and efficient parameter updates. Additionally, in convolutional neural networks (CNNs), backpropagation through convolution layers requires specific mathematical operations to compute gradients with respect to both inputs and weights.
The Cross-Entropy Loss Function [9] penalises the model when the predicted probability for the true class is low. If the model predicts a high probability for the correct class, the loss becomes small, indicating good performance (see Eq. 19).

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{19}$$

Where y_i: the true label (one-hot encoded), where only the correct class has value 1. $\hat{y}_i$: predicted probability of class i from the Softmax output. C: number of classes.
The Gradient Descent Weight [18] update rule shows the direction in which the loss increases. Subtracting the gradient moves the weights in the direction of decreasing loss (see Eq. 20).

$$W_{\text{new}} = W_{\text{old}} - \eta \frac{\partial L}{\partial W} \tag{20}$$

Where W_{old}: current weights, W_{new}: updated weights, η: learning rate (step size), and $\frac{\partial L}{\partial W}$: gradient of loss w.r.t. weights.
Gradient w. r. t. input (Backpropagation Through Convolution) [19] reversing the forward convolution operation. Rotating the kernel ensures that gradient flow follows the same spatial relationship as the original convolution (see Eq. 21).

$$\frac{\partial L}{\partial I} = \frac{\partial L}{\partial O} * rot180(K) \tag{21}$$

Where I: input image, O: output of convolution, K: convolution kernel, $\text{rot}180\ (K)$: kernel rotated by 180 degrees, and $*$: convolution operation.
The Adam optimiser [11] update helps the optimiser smooth out noisy gradients and follow consistent directions (see Eq. 22).

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{22}$$

Where g_t: gradient at time t, m_t: moving average of gradients, and β_1: momentum decay rate (typically 0.9).
Second Moment Estimate (RMS Component) estimates the variance of the gradients and helps control the step size for each parameter (see Eq. 23).

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \tag{23}$$

Where v_t: moving average of squared gradients, and β_2: decay rate for squared gradients (usually 0.999).
Adam parameter updates adaptively scales the learning rate for each parameter large steps when gradients are stable, and small steps when gradients fluctuate (see Eq. 24).

$$\theta_t = \theta_{t-1} - \eta \frac{m_t}{\sqrt{v_t} + \epsilon} \tag{24}$$

Where θ_t: updated parameter, η: learning rate, m_t: momentum term, v_t: variance term, and ϵ: small constant for numerical stability.

3.4 Performance Analysis

The performance of a classification model is commonly evaluated using the confusion matrix, which précises prediction outcomes across binary. It provides a rich foundation

for evaluating classification models. Metrics such as accuracy, precision, recall, F1-score, and AUC offer complementary insights into model behaviour (see Eq 25-29)[5].

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \tag{25}$$

$$\text{Precision} = \frac{TP}{TP+FP} \tag{26}$$

$$\text{Recall} = \frac{TP}{TP+FN} \tag{27}$$

$$F_1 = 2\frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}} \tag{28}$$

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\ d(\text{FPR}) \tag{29}$$

Where True Positive (TP) refers to correctly predicted positive cases, while False Positive (FP) represents negative samples incorrectly classified as positive. True Negative (TN) indicates correctly predicted negative cases, and False Negative (FN) refers to positive samples incorrectly predicted as negative. Using these values, the True Positive Rate (TPR), also known as Recall or Sensitivity, is computed as $\text{TPR} = \frac{TP}{TP+FN}$, measuring the proportion of actual positives correctly identified by the model. Similarly, the False Positive Rate (FPR), given by $\text{FPR} = \frac{FP}{FP+TN}$, represents the proportion of negative samples that are mistakenly classified as positive. These metrics form the foundation of performance evaluation in classification models [8].

4 Results and Discussion

This research begins by developing Convolutional Neural Network (CNN) models with one, two, and three convolutional layers, using the Softmax activation function at the output layer. During training, it was observed that the training accuracy increased rapidly and the training loss decreased sharply; however, the validation accuracy and validation loss did not show similar improvement, indicating clear signs of overfitting. To address this issue, the study introduced various data augmentation techniques and replaced the Softmax activation with the Sigmoid activation function. These modifications resulted in improved accuracy, a more stable model, and a significant reduction in overfitting.

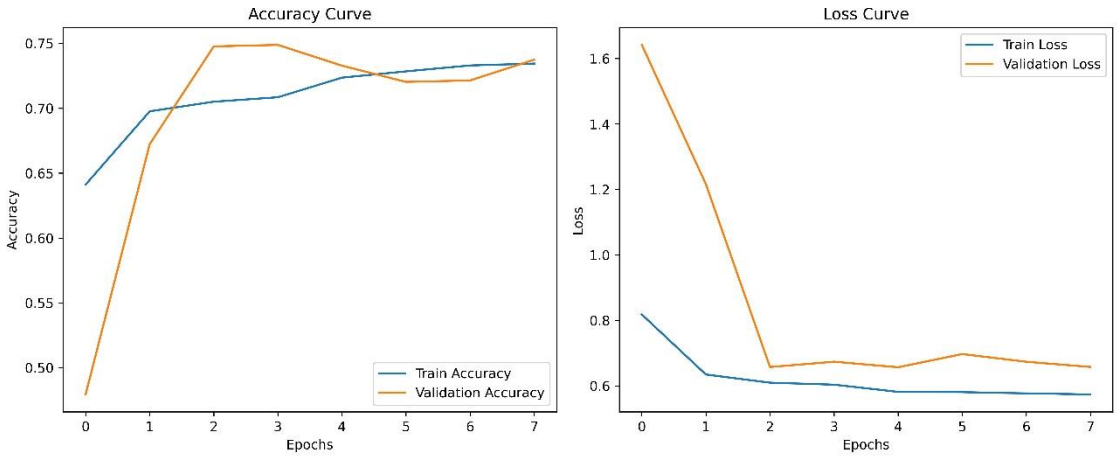

Fig. 2. Training and Validation Accuracy–Loss Curves of the CNN model with two layers and a Sigmoid output function.

Fig. 2 illustrates the training and validation performance of the CNN model across seven epochs. In the Accuracy Curve (left), the training accuracy shows a steady upward trend, increasing consistently with each epoch. The validation accuracy also improves initially and closely follows the training accuracy, indicating that the model is learning meaningful patterns from the data. However, minor fluctuations are observed after the third epoch, suggesting slight instability in generalisation, though overall performance remains comparable to the training curve. In the Loss Curve (right), the training loss decreases rapidly during the first few epochs and stabilises at a lower value, demonstrating effective minimisation of error during training. The validation loss also decreases substantially at the beginning, but after the third epoch, it shows small fluctuations instead of further improvement. This behaviour suggests that while the model generalises reasonably well, there are early signs of mild overfitting as the training loss continues to decrease, but the validation loss does not follow the same consistent downward pattern. Overall, the curves indicate that the model is learning effectively and is a stable model.

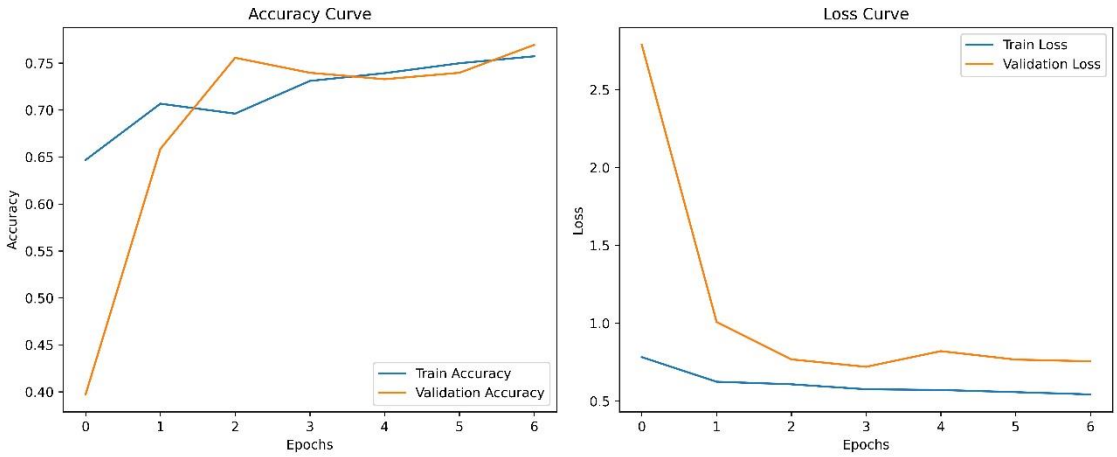

Fig. 3. Training and Validation Accuracy–Loss Curves of the CNN model with three layers and a Sigmoid output function.

Fig. 3 shows that the model exhibits stable and consistent learning behaviour as reflected in both the accuracy and loss curves. The training accuracy steadily increases across epochs, while the validation accuracy initially rises rapidly and later stabilises at a level comparable to the training accuracy, indicating good generalisation. The loss curves further support this pattern, with both training and validation loss decreasing significantly over the epochs. Although a minor fluctuation appears in the validation loss around the middle epochs, it quickly stabilizes, suggesting only mild variance rather than overfitting. Overall, it demonstrates that the model is learning effectively, maintaining a balanced performance across both training and validation datasets.

Table 3 shows the comparative performance analysis of all CNN models, both with and without augmentation, highlights a consistent improvement in classification capability as the model architecture becomes deeper. CNN1 demonstrates the weakest performance, achieving an overall accuracy of 0.64 and an AUC of 0.70, indicating limited ability to generalize across the two classes. CNN2 shows a substantial improvement, reaching 0.71 accuracy and 0.80 AUC, reflecting better feature extraction and more stable learning. The best results among all models are obtained by CNN3, which achieves 0.82 accuracy, a high macro F1-score of 0.82, and the highest AUC of 0.88, demonstrating strong discriminatory power between the Crowded and Uncrowded classes.

Table 3. Performance Comparison of CNN Models with and without Data Augmentation.

Model	**Class**	**Precision**	**Recall**	**F1-Score**	**Support**	**Overall Accuracy**	**Macro Avg F1**	**AUC**
CNN1	Crowded	0.75	0.62	0.68	528	**0.64**	**0.64**	**0.70**

Model	Class	Precision	Recall	F1-Score	Support	Overall Accuracy	Macro Avg F1	AUC
	Uncrowded	0.54	0.69	0.61	348			
CNN2	Crowded	0.80	0.70	0.75	528	**0.71**	**0.71**	**0.80**
	Uncrowded	0.62	0.74	0.67	348			
CNN3	Crowded	0.87	0.83	0.85	529	**0.82**	**0.82**	**0.88**
	Uncrowded	0.76	0.82	0.79	348			
CNN2 + Augmentation	Crowded	0.74	0.87	0.80	528	**0.73**	**0.71**	**0.79**
	Uncrowded	0.72	0.53	0.61	348			
CNN3 + Augmentation	Crowded	0.75	0.85	0.80	528	**0.74**	**0.72**	**0.82**
	Uncrowded	0.72	0.58	0.64	348			

When augmentation is applied, CNN2 benefits from enhanced data variability and improves to 0.73 accuracy, although the macro F1 remains at 0.71 due to lower recall for the Uncrowded class. CNN3 with augmentation also maintains strong performance, achieving 0.74 accuracy and 0.82 AUC, confirming that augmentation increases robustness but slightly reduces overall accuracy compared to the non-augmented CNN3. Overall, the analysis concludes that CNN3 without augmentation provides the best overall performance, while augmentation primarily improves class-balance robustness and generalisation for CNN2 and CNN3, especially in detecting the Crowded category. The results show that the baseline models CNN1, CNN2, and CNN3 achieve progressively better performance, with CNN3 attaining the highest overall accuracy (0.82), macro F1-score (0.82), and AUC (0.88). However, despite their high accuracy, these three models exhibit signs of overfitting, as indicated by the imbalance between class-wise recall values and their reliance on training-specific patterns. To address this issue, data augmentation techniques were incorporated into CNN2 and CNN3, producing more generalised models. CNN2 + Augmentation improves the recall for Crowded images (0.87) and maintains competitive overall performance (accuracy = 0.73, AUC = 0.79), while CNN3 + Augmentation achieves a more balanced trade-off between precision and recall across classes (accuracy = 0.74, AUC = 0.82). Additionally, the Sigmoid activation function in the output layer was used to stabilise the probability

estimation for binary classification, improving generalisation. Overall, the CNN with three layers augmentation and sigmoid activation function demonstrate better robustness compared to their overfitted counterparts.

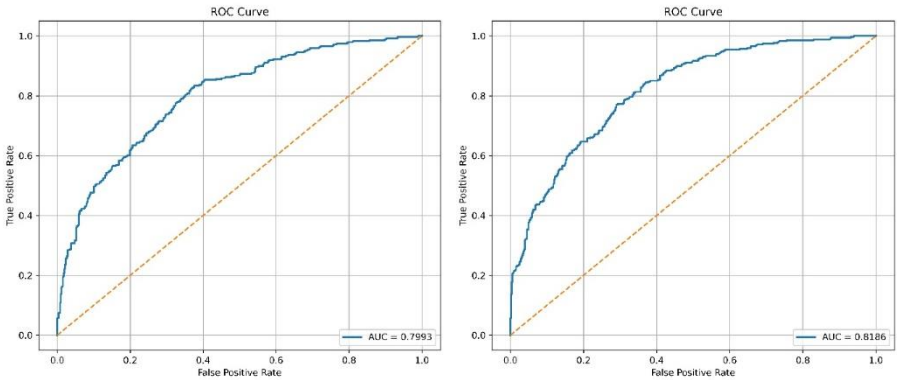

Fig. 4. Comparative ROC Curves of CNN Models with two and three layers for Crowded vs. Uncrowded Image Classification.

Fig. 4 illustrates the ROC curves, which collectively show the classification performance of the models across different decision thresholds. Both curves rise well above the diagonal reference line, representing random guessing. The first model achieves an AUC of 0.7993, demonstrating a strong ability to discriminate between the Crowded and Uncrowded classes, even though the curve is slightly less steep than higher-performing models. The second model performs even better, with an AUC of approximately 0.82, indicating a higher reliability in separating the two classes. In both cases, the smooth upward progression toward the upper-left corner reflects good sensitivity at low false-positive rates and a solid balance between true-positive and false-positive rates across thresholds. Overall, the ROC curves confirm that both models exhibit reliable classification capability, with the second model showing superior performance and making it more suitable for practical crowd-classification applications.

5 Conclusion

A comprehensive performance comparison across all models further highlights these findings. CNN3 achieves the highest accuracy (0.82), macro F1-score (0.82), and AUC (0.88) among the non-augmented models, demonstrating superior discriminatory capability between Crowded and Uncrowded classes. With augmentation, both CNN2 and CNN3 show improved robustness, especially in detecting the Crowded class, with CNN2 + Augmentation achieving a recall of 0.87 and CNN3 + Augmentation maintaining a strong AUC score of 0.82. Although augmentation slightly reduces the peak accuracy of CNN3, it enhances class balance and generalisation, particularly where the dataset exhibits variability. The ROC curve analysis further reinforces these conclusions. Both the two-layer and three-layer Sigmoid-based models achieve AUC values

of 0.7993 and 0.82 respectively, confirming reliable classification performance well above random guessing. The smoother and steeper rise of the three-layer model reflects its superior ability to distinguish between the two classes, making it more suitable for real-world crowd-classification scenarios. Overall, the results demonstrate that the three-layer CNN with Sigmoid activation provides the strongest baseline performance, while the augmented versions of CNN2 and CNN3 deliver enhanced generalisation and class-balanced robustness. The combination of deeper architecture, Sigmoid activation, and targeted augmentation yields a stable and effective model suitable for practical Crowded vs. Uncrowded image classification applications.

References

1. Madani, A. A., Banunnah, S. M., Serri, L. Z., Ghalib, R. A., & Yaghmour, S. M. (2025). Enhancing Risk Management in Events by Providing AI-Risk Assessment Framework. International Journal of Research and Studies Publishing, 6(71), 142-178.
2. Alafif, T., Jassas, M., Abdel-Hakim, A. E., Alfattni, G., Althobaiti, H., Ikram, M., ... & Khayyat, K. (2025). Towards an Integrated Intelligent Framework for Crowd Control and Management (IICCM). IEEE Access.
3. Alasmari, A. M., Farooqi, N. S., & Alotaibi, Y. A. (2024). Recent trends in crowd management using deep learning techniques: a systematic literature review. Journal of Umm Al-Qura University for Engineering and Architecture, 1-29.
4. Raj Kumar Sharma, Mukta Sharma. (2024). Garbage Bin Classification Using Convolutional Neural Networks with Clean and Noisy Bin Images. Journal of Computational Analysis and Applications (JoCAAA), 33(08), 3053–3068. https://eudoxuspress.com/index.php/pub/article/view/2406.
5. Sharma, R.K., Sangeeta (2024). Fractured Bone Diagnostics Using Optimised Convolutional Neural Network Model with Variable Learning Rates. In: Alam, M.A., Siddiqui, F., Zafar, S., Hussain, I. (eds) Proceedings of 4th International Conference on ICT for Digital, Smart, and Sustainable Development. ICIDSSD 2024. Innovations in Sustainable Technologies and Computing. Springer, Singapore. https://doi.org/10.1007/978-981-97-7831-7_15.
6. Macriga, G. A., Bavyatha, S., Abhinidhi, S. H., & Jagadeesh, N. (2024, October). Crowd Management Using AI & ML. In 2024 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS) (pp. 1-5). IEEE.
7. H. R, M. Thanawala, S. Shukla, H. Upadhyay, S. Atharabbas and V. Chella, "IoT Based Crowd Detection and Stampede Avoidance using Predictive Analysis," 2023 International Conference on Artificial Intelligence and Knowledge Discovery in Concurrent Engineering (ICECONF), Chennai, India, 2023, pp. 1-7, doi: 10.1109/ICECONF57129.2023.10084059.
8. Sharma, R. K., & Jailia, M. (2023). IoT based smart bin system using multilayer neural network classifier for smart cities. RECENT ADVANCES IN SCIENCES, ENGINEERING, INFORMATION TECHNOLOGY & MANAGEMENT, 2782(1), 020019. https://doi.org/10.1063/5.0154169.
9. Mao, A., Mohri, M., & Zhong, Y. (2023, July). Cross-entropy loss functions: Theoretical analysis and applications. In *International conference on Machine learning* (pp. 23803-23828).
10. Sharma, Raj Kumar and Mogha, Naina and Saini, Muskan and Kumar, Ashish, Garbage Bin Status Indicator Based on Multilayer Convolutional Neural Networks (May 5, 2023). Proceedings of the KILBY 100 7th International Conference on Computing Sciences 2023

(ICCS 2023), SSRN: https://ssrn.com/abstract=4489022 or http://dx.doi.org/10.2139/ssrn.4489022.

11. Shahade, A. K., Walse, K. H., & Thakare, V. M. (2023). Deep learning approach-based hybrid fine-tuned Smith algorithm with Adam optimiser for multilingual opinion mining. *International Journal of Computer Applications in Technology*, *73*(1), 50-65.
12. Suthar, M., Sharma, S., & Kumar, S. (2023, November). Current status and future directions for crowd management using machine learning. In Proceedings of the 5th International Conference on Information Management & Machine Intelligence (pp. 1-5).
13. Tyagi, B., Nigam, S., & Singh, R. (2022). A review of deep learning techniques for crowd behaviour analysis. Archives of Computational Methods in Engineering, 29(7), 5427-5455.
14. Almutairi, M. M., Yamin, M., Halikias, G., & Abi Sen, A. A. (2021). A framework for crowd management during COVID-19 with artificial intelligence. Sustainability, 14(1), 303.
15. S. Islam, A. Kafi, M. Z. Islam, N. Islam and M. N. Ullah, "IoT Based Crowd Congestion and Stampede Avoidance in Hajj Using Wemos D1 with Machine Learning Approach," 2019 4th International Conference on Electrical Information and Communication Technology (EICT), Khulna, Bangladesh, 2019, pp. 1-5, doi: 10.1109/EICT48899.2019.9068814.
16. Haque, S., Sadi, M. S., Rafi, M. E. H., Islam, M. M., & Hasan, M. K. (2019, July). Real-time crowd detection to prevent stampede. In Proceedings of International Joint Conference on Computational Intelligence: IJCCI 2018 (pp. 665-678). Singapore: Springer Nature Singapore.
17. Zhao, L., Wang, J., Liu, J., & Kato, N. (2019). Routing for crowd management in smart cities: A deep reinforcement learning perspective. IEEE Communications Magazine, 57(4), 88-93.
18. Sum, J., Leung, C. S., & Ho, K. (2019). A limitation of gradient descent learning. *IEEE Transactions on Neural Networks and Learning Systems*, *31*(6), 2227-2232.
19. Murugan, P. (2017). Feed forward and backward run in deep convolution neural network. *arXiv preprint arXiv:1711.03278*.
20. Simon, B. I., Von Hagen, S., Deasy, M. J., Faldu, M., & Resnansky, D. (2000). Changes in alveolar bone height and width following ridge augmentation using bone graft and membranes. *Journal of Periodontology*, *71*(11), 1774-1791.